

```
tsvc/s112 · s112.c · bare_llm · claude-haiku-4-5-20251001 · parallel-not-faster
```

BEFORE

132 }

133

```
134 static real_t kernel_s112(void)
```

135 {

```
136     for (int nl = 0; nl < R; nl++) {
```

```
137     for (int i = LEN 1D - 2; i >= 0; i--) {
```

```
138         a[i+1] = a[i] + b[i];
```

139 }

```
140      pb mix(nl);
```

141 }

```
142     return (real t)0;
```

143 }

144

AFTER

132 }

133

```
134 static real_t kernel_s112(void)
```

135 {

```
136     for (int nl = 0; nl < R; nl++) {
```

```
140     real_t* a_copy = (real_t*)malloc(LEN_1D * sizeof(real_t));
```

```
141     memcpy(a_copy, a, LEN_1D * sizeof(real_t));
```

142

```
143      /* Inner loop can be parallelized: each iteration i writes to a[i+1],
```

```
144         reading from a copy[i] and b[i]. No two iterations write to the
```

```
145         same element, so there are no data races. */
```

```
146      #pragma omp parallel for
```

```
147     for (int i = LEN 1D - 2; i >= 0; i--) {
```

```
148         a[i+1] = a_copy[i] + b[i];
```

149 }

```
151         free(a copy);
```

```
152         pb mix(nl);
```

153 }

```
154     return (real t)0;
```

155 }

156

file